

Smart-Contract Vulnerability Detection — Our Tool vs. Baselines

Per-dataset, per-vulnerability detection metrics, recomputed uniformly from each tool's own TP/FP/TN/FN. **Analyzed** = TP+FP+TN+FN; **Total** = contracts in that dataset–vulnerability benchmark; **Failures** = Total – Analyzed = contracts the tool could not analyse (compile error / timeout / crash), excluded from the metrics. **NA** = tool has no detector for / was not run on that case.

RSD (Ressi et al.)

► Reentrancy

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	66	23	49	0	138	138	0	74.2	100.0	85.2	25.8	0.0
Slither	62	38	34	4	138	138	0	62.0	93.9	74.7	38.0	6.1
SmartCheck	5	12	60	61	138	138	0	29.4	7.6	12.0	70.6	92.4
Sailfish	NA	NA	NA	NA	NA	138	NA	NA	NA	NA	NA	NA
Mythril	56	44	26	10	136	138	2	56.0	84.9	67.5	44.0	15.2
Oyente+	41	30	41	24	136	138	2	57.8	63.1	60.3	42.2	36.9
Osiris	NA	NA	NA	NA	NA	138	NA	NA	NA	NA	NA	NA
Vandal	61	51	19	4	135	138	3	54.5	93.8	68.9	45.5	6.2
EtherSolve	0	0	72	66	138	138	0	0.0	0.0	0.0	0.0	100.0

SBC (SmartBugs Curated)

► Reentrancy

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	30	8	100	0	138	142	4	79.0	100.0	88.2	21.1	0.0
Slither	29	28	78	0	135	142	7	50.9	100.0	67.4	49.1	0.0
SmartCheck	29	56	55	2	142	142	0	34.1	93.5	50.0	65.9	6.5
Sailfish	23	4	104	8	139	142	3	85.2	74.2	79.3	14.8	25.8
Mythril	29	39	72	2	142	142	0	42.6	93.5	58.6	57.4	6.5
Oyente+	29	9	101	2	141	142	1	76.3	93.5	84.1	23.7	6.5
Osiris	26	23	68	0	117	142	25	53.1	100.0	69.3	46.9	0.0
Vandal	20	48	63	11	142	142	0	29.4	64.5	40.4	70.6	35.5
EtherSolve	30	56	54	1	141	142	1	34.9	96.8	51.3	65.1	3.2

► Overflow

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	15	25	98	0	138	142	4	37.5	100.0	54.5	62.5	0.0
Slither	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
SmartCheck	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
Sailfish	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
Mythril	12	18	109	3	142	142	0	40.0	80.0	53.3	60.0	20.0
Oyente+	13	101	25	2	141	142	1	11.4	86.7	20.2	88.6	13.3
Osiris	13	59	43	2	117	142	25	18.1	86.7	29.9	81.9	13.3
Vandal	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
EtherSolve	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA

► Timestamp

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	14	9	115	0	138	142	4	60.9	100.0	75.7	39.1	0.0
Slither	14	9	112	0	135	142	7	60.9	100.0	75.7	39.1	0.0
SmartCheck	1	0	128	13	142	142	0	100.0	7.1	13.3	0.0	92.9
Sailfish	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
Mythril	12	2	126	2	142	142	0	85.7	85.7	85.7	14.3	14.3
Oyente+	11	2	125	3	141	142	1	84.6	78.6	81.5	15.4	21.4
Osiris	4	1	103	9	117	142	25	80.0	30.8	44.4	20.0	69.2
Vandal	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
EtherSolve	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA

► Tod

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	4	15	119	0	138	142	4	21.1	100.0	34.8	79.0	0.0
Slither	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
SmartCheck	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
Sailfish	2	62	73	2	139	142	3	3.1	50.0	5.9	96.9	50.0
Mythril	1	37	101	3	142	142	0	2.6	25.0	4.8	97.4	75.0
Oyente+	2	35	102	2	141	142	1	5.4	50.0	9.8	94.6	50.0
Osiris	2	29	86	0	117	142	25	6.5	100.0	12.1	93.5	0.0
Vandal	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA
EtherSolve	NA	NA	NA	NA	NA	142	NA	NA	NA	NA	NA	NA

Qian

► Reentrancy

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	69	11	142	0	222	222	0	86.2	100.0	92.6	13.8	0.0
Slither	68	12	140	1	221	222	1	85.0	98.6	91.3	15.0	1.5
SmartCheck	62	110	43	7	222	222	0	36.0	89.9	51.4	63.9	10.1
Sailfish	46	2	151	23	222	222	0	95.8	66.7	78.6	4.2	33.3
Mythril	36	60	93	33	222	222	0	37.5	52.2	43.6	62.5	47.8
Oyente+	61	73	80	8	222	222	0	45.5	88.4	60.1	54.5	11.6
Osiris	64	139	14	5	222	222	0	31.5	92.8	47.1	68.5	7.2
Vandal	62	137	16	7	222	222	0	31.2	89.9	46.3	68.8	10.1
EtherSolve	65	82	71	4	222	222	0	44.2	94.2	60.2	55.8	5.8

► Overflow

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	86	9	180	0	275	275	0	90.5	100.0	95.0	9.5	0.0
Slither	NA	NA	NA	NA	NA	275	NA	NA	NA	NA	NA	NA
SmartCheck	NA	NA	NA	NA	NA	275	NA	NA	NA	NA	NA	NA
Sailfish	NA	NA	NA	NA	NA	275	NA	NA	NA	NA	NA	NA
Mythril	23	2	187	63	275	275	0	92.0	26.7	41.4	8.0	73.3
Oyente+	63	46	143	23	275	275	0	57.8	73.3	64.6	42.2	26.7
Osiris	52	19	168	34	273	275	2	73.2	60.5	66.2	26.8	39.5
Vandal	NA	NA	NA	NA	NA	275	NA	NA	NA	NA	NA	NA
EtherSolve	NA	NA	NA	NA	NA	275	NA	NA	NA	NA	NA	NA

► Timestamp

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	173	29	147	0	349	349	0	85.6	100.0	92.3	14.4	0.0
Slither	110	117	59	63	349	349	0	48.5	63.6	55.0	51.5	36.4
SmartCheck	2	1	175	171	349	349	0	66.7	1.2	2.3	33.3	98.8
Sailfish	NA	NA	NA	NA	NA	349	NA	NA	NA	NA	NA	NA
Mythril	59	86	90	114	349	349	0	40.7	34.1	37.1	59.3	65.9
Oyente+	1	15	160	172	348	349	1	6.2	0.6	1.1	93.8	99.4
Osiris	2	16	157	164	339	349	10	11.1	1.2	2.2	88.9	98.8
Vandal	NA	NA	NA	NA	NA	349	NA	NA	NA	NA	NA	NA
EtherSolve	NA	NA	NA	NA	NA	349	NA	NA	NA	NA	NA	NA

SolidiFI

► Tod

Tool	TP	FP	TN	FN	Analyzed	Total	Failures	Precision %	Recall %	F1 %	FDR %	FNR %
Our Tool (ours)	50	23	27	0	100	100	0	68.5	100.0	81.3	31.5	0.0
Slither	NA	NA	NA	NA	NA	100	NA	NA	NA	NA	NA	NA
SmartCheck	NA	NA	NA	NA	NA	100	NA	NA	NA	NA	NA	NA
Sailfish	40	5	43	1	89	100	11	88.9	97.6	93.0	11.1	2.4
Mythril	21	0	50	29	100	100	0	100.0	42.0	59.2	0.0	58.0
Oyente+	36	4	46	14	100	100	0	90.0	72.0	80.0	10.0	28.0
Osiris	NA	NA	NA	NA	NA	100	NA	NA	NA	NA	NA	NA
Vandal	NA	NA	NA	NA	NA	100	NA	NA	NA	NA	NA	NA
EtherSolve	NA	NA	NA	NA	NA	100	NA	NA	NA	NA	NA	NA

Our Tool (ours) best F1 per block Failures = contracts the tool could not analyse (N/A)